

PROJECT REPORT

Enterprise Corporate Network Infrastructure Using MikroTik & Cisco Technologies

An enterprise-style corporate network lab designed, implemented, tested, and documented in EVE-NG.

Project Type	Enterprise Network Infrastructure Lab
Network Emulator	EVE-NG
Routing Platform	MikroTik RouterOS v7
Switching Platform	Cisco Layer 2 Switching
Primary Technologies	BGP, VLAN, LACP, DHCP, NAT, Firewall, IPsec, GRE
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1. Executive Summary

This project demonstrates the design, implementation, and validation of an enterprise-style corporate network built in EVE-NG using MikroTik RouterOS v7 and Cisco Layer 2 switching technologies. The lab represents a headquarters environment with dual ISP connectivity, campus switching, VLAN segmentation, centralized network services, and two remote branch offices.

The project focuses on practical enterprise networking skills rather than isolated configuration exercises. Routing, switching, WAN connectivity, VPNs, security policies, link aggregation, and troubleshooting were integrated into a single working environment.

2. Project Objectives

- Design a scalable enterprise campus network architecture.
- Integrate MikroTik routing with Cisco Layer 2 switching.
- Implement dual-ISP connectivity using BGP.
- Segment the enterprise LAN using IEEE 802.1Q VLANs.
- Provide inter-VLAN connectivity and centralized DHCP services.
- Implement LACP link aggregation for bandwidth and link redundancy.
- Connect remote branches using IPsec Site-to-Site VPN and GRE.
- Apply NAT and firewall filtering to control network traffic.
- Validate connectivity through structured testing and troubleshooting.
- Document the implementation in a professional, reusable format.

3. Network Architecture

The architecture is organized around a headquarters/core network and two remote branches. The headquarters router provides WAN connectivity and routing functions, while Cisco switches provide Layer 2 campus connectivity to multiple floors/access segments.

Component	Role
ISP-1	Primary/alternate upstream Internet connectivity using eBGP
ISP-2	Secondary upstream Internet connectivity using eBGP
HQ Core Router	MikroTik RouterOS v7; WAN routing, BGP, NAT, firewall, VPN and routing functions
Access Router	MikroTik RouterOS v7; enterprise access/routing functions and internal connectivity
Cisco Core Switch	Layer 2 campus aggregation and VLAN/trunk connectivity
Floor Access Switches	User/device access and uplink connectivity using LACP where implemented
Branch 1	Remote site connected using IPsec Site-to-Site VPN
Branch 2	Remote site connected using GRE tunnel

4. Device and Software Environment

Item	Implementation
Emulation	EVE-NG
HQ Router	MikroTik RouterOS v7.8
Access Router	MikroTik RouterOS v7.16

Item	Implementation
Switching	Cisco Layer 2 switches
Configuration Method	CLI / RouterOS and Cisco IOS configuration
Testing	Ping, traceroute, routing/BGP/VPN/LACP status verification

5. Technologies Implemented

5.1 Routing and WAN

- eBGP for dual-ISP connectivity.
- iBGP where applicable for internal BGP communication.
- Static routing for selected internal/tunnel paths.
- BGP route filtering to control route exchange.
- IP address planning and management.

5.2 Switching

- IEEE 802.1Q VLANs.
- VLAN trunking and access-port configuration.
- Layer 2 traffic forwarding.
- LACP / IEEE 802.3ad link aggregation.
- Multi-floor campus switching.

5.3 Network Services and Security

- DHCP server configuration.
- NAT for Internet access.
- Firewall filter rules.
- Controlled routing between network segments.

5.4 Branch Connectivity

- IPsec Site-to-Site VPN for Branch 1.
- GRE tunnel for Branch 2.
- Static/tunnel routing as required by the design.

6. Topology Documentation

The project is documented through multiple topology views so that different technical aspects can be understood independently.

Diagram	Purpose
Enterprise Topology	Complete physical/logical representation of HQ, ISPs, switches, and branches.
Logical Topology	Shows Layer 3 relationships and major traffic/routing paths.
VLAN Topology	Shows VLAN segmentation, trunks, access networks, and inter-VLAN connectivity.
BGP Topology	Shows ISP AS relationships, eBGP/iBGP sessions, and routing redundancy.
VPN Topology	Shows IPsec and GRE branch connectivity and protected/tunneled paths.

7. Implementation Summary

7.1 Dual ISP and BGP

Two upstream providers were connected to the headquarters router. eBGP sessions were configured to exchange routes, with route filtering applied to control advertisements and improve routing policy.

7.2 VLAN and Inter-VLAN Routing

The campus LAN was segmented using IEEE 802.1Q VLANs. Trunks were used between routing and switching components, while access ports connected endpoint networks. Inter-VLAN routing was implemented at the routing layer.

7.3 LACP

LACP bonding/EtherChannel was implemented between supported MikroTik and Cisco devices to combine physical links. This provides increased aggregate capacity and improves resilience against an individual link failure.

7.4 DHCP, NAT and Firewall

DHCP services provide client addressing. NAT provides Internet access where required, while firewall filter rules control traffic according to the security policy of the lab.

7.5 Branch 1 IPsec

Branch 1 was connected to headquarters using a Site-to-Site IPsec VPN. The tunnel provides encrypted communication between the defined headquarters and branch networks.

7.6 Branch 2 GRE

Branch 2 was connected using a GRE tunnel. Tunnel endpoints and routing were configured so that the remote network could communicate through the logical GRE interface.

8. Verification and Testing

Validation was performed after implementation to confirm that each major service operated as designed. Verification evidence should be stored in the repository under the Verification directory.

Area	Verification Evidence
BGP	BGP neighbor/session status, received/advertised routes, routing table
VLAN	VLAN database, trunk status, access-port assignment, endpoint connectivity
LACP	Bond/Port-Channel status and member-link state
DHCP	Active leases and client IP allocation
NAT	NAT/connection tracking and successful Internet reachability
Firewall	Rule counters/logs and permitted/blocked traffic tests
IPsec	IKE/IPsec SA status and protected-network connectivity
GRE	Tunnel interface status and end-to-end reachability
Routing	Routing table, ping and traceroute results

9. Troubleshooting Methodology

The lab was also used to practice structured enterprise troubleshooting. Problems were approached from the physical/link layer toward Layer 2, Layer 3, routing, VPN, and policy layers.

- Check interface/link status and physical/logical connectivity.

- Verify VLAN membership, trunks, and LACP member links.
- Check IP addressing, gateways, and ARP/neighbor information.
- Review routing tables and next-hop reachability.
- Verify BGP neighbor state and route advertisements.
- Check VPN tunnel and security association status.
- Review firewall/NAT rules and counters.
- Use ping and traceroute to isolate the failing path.
- Document the cause, corrective action, and verification result.

10. Security Considerations

- Use firewall filtering to restrict unnecessary traffic.
- Use encrypted IPsec tunnels for sensitive branch-to-HQ communication.
- Apply route filtering to control BGP advertisements.
- Separate enterprise departments and services using VLANs.
- Restrict administrative access to network infrastructure.
- Use strong credentials and secure management protocols in production deployments.

11. Key Learning Outcomes

- Enterprise network architecture and structured design.
- MikroTik RouterOS v7 routing and security configuration.
- Cisco Layer 2 switching and EtherChannel.
- BGP-based WAN/ISP connectivity and route policy.
- VLAN segmentation and inter-VLAN routing.
- IPsec and GRE-based branch connectivity.
- Network troubleshooting across multiple layers.
- Professional network documentation and verification.

12. Future Improvements

- OSPF multi-area routing.
- IS-IS routing.
- VRRP-based gateway high availability.
- MPLS and MPLS VPN.
- FortiGate firewall integration.
- RADIUS/AAA authentication.
- SNMP-based network monitoring.
- Syslog and centralized logging.
- NetFlow/traffic analysis.
- IPv6 deployment.
- Zero Trust network security architecture.

13. GitHub Repository Structure

```
Enterprise-Corporate-Network/  
  README.md  
  Topology/  
  01-Enterprise-Architecture/  
  02-Logical-Topology/  
  03-VLAN-Architecture/  
  04-BGP-Architecture/  
  05-VPN-Architecture/  
  Configurations/  
  Routing/  
  VPN/  
  VLAN/  
  DHCP/  
  Firewall/  
  NAT/  
  LACP/  
  Verification/  
  Troubleshooting/  
  Documents/  
  Project_Report.pdf  
  IP_Addressings.xlsx  
  Lessons_Learned.md
```

14. Conclusion

This project provided a practical environment for implementing an enterprise corporate network using MikroTik and Cisco technologies. By combining dual-ISP BGP connectivity, VLAN-based campus switching, LACP, DHCP, NAT, firewall filtering, IPsec, and GRE, the lab demonstrates how multiple networking technologies work together as a complete infrastructure.

Beyond configuration, the project emphasized validation, troubleshooting, documentation, and operational thinking. These are essential skills for enterprise network engineering and provide a strong foundation for further work in network security and cloud networking.

15. Author

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Project focus: Enterprise Networking • MikroTik • Cisco • Network Security • Cloud Networking